MIL-W-50577C (AR)
AMENDMENT 4
17June 1983
SUPERSEDING
AMENDMENT 3
06 August 1982

MILITARY SPECIFICATION

WARHEAD, 155MM, M549A1
BONDING ASSEMBLY

This amendment forms a part of Military Specification MIL-W-50577C (AR), dated 20 June 1979 and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* 2.2: Delete ASTM "E18" and substitute:

"E103 Rapid Indentaton Hardness Testing of Metallic Materials".

PAGE 5

* 3.11, Delete: "25KSI In. (Information only)." and substitute "25 KSI In."

PAGE 8

* "Table 1" Under "Warhead Unpainted Metal Parts", delete: "Interior Surface Test....4.5.6" and substitute: "Precracked Charpy Test 1 100% 3.11 4.5.4.3/Appendix".

PAGE 12

* 4.4.2.3: Delete Critical defect 1 in its entirety. Add Special defect A as follows:

"Special
A Precracked charpy Test 4.4.3.1.4 3.11 4.5.4.3/Appendix"

PAGE 13

* 4.4.2.3: Delete Major defect 122 in its entirety.

PAGE 17

* 4.4.2.7: Major defect 101, delete "100%" and substitute: "0.40%".

FSC 1320

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* 4.4.3.1.1: Delete, "The test shall be conducted...samples of paragraph 4.4.3.1.2."

* Add 4.4.3.1.4 as follows:

"4.4.3.1.4 Slow bend precracked charpy test. Beginning with the first heat treat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated warhead from each heat treated lot shall be subjected to the precrack charpy test. The warhead selected shall be of the highest average hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested, with acceptance based on the average crack strength. In addition, no individual crack strength shall be less than 23 KSI $\sqrt{in.}$ If one test is invalid, acceptance may be based on the crack strength of the valid sample, or another sample from the same warhead may be tested. If either the average or individual crack strength is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to retemper and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be re-instituted, until three consecutive heat treated lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government.

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PAGE 19

Delete paragraph 4.4.3.4 in its entirety and substitute the following:

"4.4.3.4 Salt Spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490".

Add 4.4.5 as follows:

"4.4.5 Steel quality. Steel supplied shall comply with MIL-S-50783, except that macroetch quality shall be per MIL-STD-1459. The quality of the steel as indicated by the macroetch results shall be equal to, or better than A5, B3 and C8 with defects D1, D3, D4, D5, D6, D7 or D8 unacceptable. Manufacturer shall provide a certification that the applicable MIL-S-50783, and macroetch requirements have been met."

PAGE 20

* 4.5.3.2: Delete "Calibration" and substitute "Verification".

PAGE 21

* 4.5.3.2: Add the following to the end of the first paragraph:

"The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.3.4)."

Delete paragraph 4.5.3.4 and substitute the following in its place:

"4.5.3.4 Operator qualifications. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonics equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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* Delete 4.5.4 in its entirety and substitute:

"4.5.4 Mechanical Properties.-

4.5.4.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. (Note: Impressions shall be completely removed by final machining.).

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The Hardness Tester shall use a standard 10MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000ks. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. One reading shall be taken at location T and one at location X as indicated on Dwg. 9235995.

4.5.4.2 Tension. The two warheads selected for testing shall have two specimens taken from each and tested as follows:

a. Base section. The base section tensile specimen shall be taken transversely, one inch from the centerline. Failure to meet the yield strength or elongation requirements of dwg. 9235995 shall require a retest from the remaining portion of the base. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results

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to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot. In addition, failure of either individual test specimen to meet 130,000 psi minimum yield strength and 3% minimum elongation shall be cause for rejection of that heat treatment inspection lot represented by the sample.

b. Side wall. One specimen shall be taken from between stations T and X. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

Testing shall be conducted in accordance with American Society for Testing and Materials, Method E8 with the largest substandard size round specimen possible.

4.5.4.3 Slow bend precrack charpy. The two specimens shall be taken from locations C1 and C2 on Figure 1. The test shall be conducted at a specimen temperature of -65 degree fahrenheit (^{O}F). The test shall be conducted in accordance with the procedure defined in the Appendix of this specification with the notches machined across the rear face of the specimen. Locations C3 and C4 on Figure 1 shall be used for replacement specimens for locations C1 and C2 respectively."

PAGE 23

* 4.5.9.1: Delete "Calibration" and substitute "Verification".

Add the following: "The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.9.3)."

PAGE 24

Delete paragraph 4.5.9.3 and substitute the following in its place:

"4.5.9.3 Operator qualifications. For operation of eddy current equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of eddy current equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

4.5.10.1: Delete this paragraph in its entirety and substitute the following:

"4.5.10.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned

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subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.4 and test samples shall be dried in accordance with Drawing 9235995."

* Delete 4.5.11 in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 1320-A709)

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